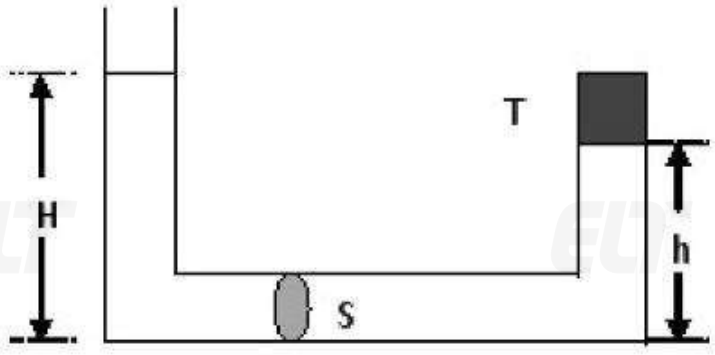
33. A pipe with constant cross–sectional area S is bent into a U–shape. One side of the pipe is open whereas the other one is closed by a cap T (as in the figure). The pipe is filled with water (density ρ) that reaches heights H ed h in the two arms. The force that the water exerts on T is equal to:



A. $\rho g(H-h)$, downwards

B. ρgh, upwards

C. ρgHS, upwards

D. $\rho g(H+h)$, downwards

E. $\rho g(H-h)S$, upwards